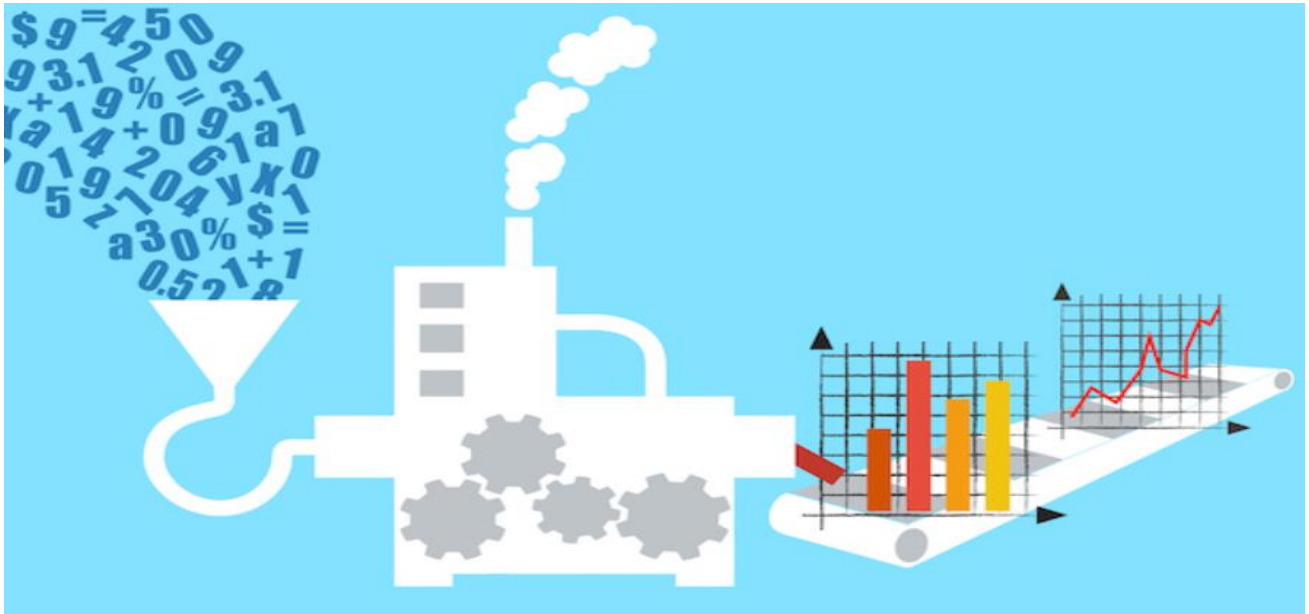




School Name: School of Computing

Academic Year: AY2627 Semester 1

Course Name: DAAA

Module Code: ST1502

Module Name: Data Visualization

Assignment: CA2 (Group Assignments)

No. of pages: 6 (including cover page), accompanied by 1 meta data and 3 data files

Deadline: 14 Aug 2026 (Friday), 8:00am

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Section 1 Instructions and Guidelines

1. This is a **group assignment** for three (3) persons, contributing **40% to the module grade**. The assignment consists of two components: a **group component**, which contributes **40% of CA2**, and an **individual component**, which contributes **60% of CA2**. The aim of this assignment is to develop charts and visualisations and derive insights using **Python Plotly Express, Python Plotly Graph Objects** and **Python Plotly Dash**.
2. Groups are to be formed at the start of the project. Every group should appoint a leader. The Group Leader is to submit the deliverable on behalf of the group and follow this filename format: "2BXX-GroupLeaderName.zip", and upload to the assignment folder at BrightSpace.
3. Below is the contents in the zip file:
 - **One Jupyter Notebook:** Source code with comments for Python Plotly, Plotly Graph Objects and Plotly Dash (other IDE or Python developer tool is not allowed)
 - **Datasets:** The datasets that you have cleaned or wrangled for analysis
 - **Powerpoint slides:** Descriptions on data wrangling, analysis of datasets, insights and recommendations
4. The zip file should include both group and individual components. Individual components submitted separately will not be marked.
5. **LATE SUBMISSION:** 50% of the marks will be deducted for assignments that are received within ONE (1) calendar day after the submission deadline. No marks will be given thereafter. Exceptions to this policy will be given to students with valid LOA on medical or compassionate grounds. In such cases, students must inform the lecturer as soon as reasonably possible. Students should not assume that an extension has been granted unless explicitly notified by the lecturer.
6. **PLAGIARISM:** No marks will be awarded, if the work is copied or you have allowed/enabled others to copy your work. Plagiarism is a serious offence, and if you are found to have committed, aided, and/or abetted the offence of plagiarism, disciplinary action will be taken against you.

Warning: Plagiarism means passing off as one's own ideas, works, writings, etc., which belong to another person. In accordance with this definition, you are committing plagiarism if you

copy the work of another person and turning it in as your own, even if you have the permission of that person.

Section 2

Scope of the assignment

Background:

The company in this use case is an IT solutions integrator. Based on clients' requirements, it proposes IT systems and supplies all necessary hardware and software. It also develops customised programs, integrates all components and deploys the integrated systems.

The company keeps the profiles of all its clients. Since 2021, it has conducted annual survey that gathers clients' feedback. For every year, it records the revenue gained from each client and the costs of hardware, software and manpower incurred (collectively known as Cost of Goods Sold, COGS). The data is kept in three datasets as attached to this assignment specifications. Detailed descriptions are found in the attached meta data.

As the data analysts of the company, your group is to write Python programs that perform data wrangling on the given datasets and present interactive data visualization that provides insights. You are to follow the requirements listed below.

1. Objective:

Define a clear objective for your analysis with the given datasets, such that it will contribute to the company with insights and recommendations.

2. Data Wrangling:

- Clean and preprocess your datasets to ensure accurate analysis.
- Document the process following the format of Table 1 sample below. It should contain the field names, number of records affected and action taken to clean the data. The sample below is an illustration, using a different use case.

Dataset name: HDB Resales Price		
Original no. of records: 32334. Duplicated rows removed: 34. Final no. of rows: 32300		
Field Name	Number of records affected	Action taken / remarks
month	0	Converted to datetime; format standardised
town	8	Cleaned abbreviations (e.g. AMK → Ang Mo Kio)
flat_type	20	Standardized (e.g. 3RM, 3Rooms → 3-Room)
...	...	...

Table 1: A summary of data quality issues and remediation actions for the given dataset

3. Plot charts:

a. **Plotly Express/Graph Objects:** Each group member is to create three (3) charts:

- One (1) chart using Plotly Express
- Two (2) charts using Plotly Graph Objects

b. **Plotly Dash:**

- Each member: Create an interactive dashboard that contains the 3 charts with dropdown lists, checkboxes, etc.
- Group: Combine all the three dashboards into a complete storyboard

4. **Insights:** Each member: explain the insights from each chart in not more than three points. Use your analysis to craft a compelling narrative.

5. **PowerPoint Slides:** Compile your process and findings into a deck of Powerpoint slides, not more than **18 slides** in total (excluding the cover page). The slides include the following:

a) **Cover Page:**

- Module Name: DAVI
- Academic year: AY2627 Sem1
- CA2: Data Analytics on IT Systems Integrator
- Student IDs and Full Names as in SAS:
e.g.:
 - P2523456 Tan Ling Ling Amanda
 - P2612345 Seah Poh Yu
 - P2624255 Ravi Krishnan

b) **Objective of the assignment (group):** Explain the objective of the assignment.

c) **Data wrangline (group):** Tabulate the process, following the example in the above Table 1.

d) **Dashboard design tempate (group):** Outline the placement for header, navigation and main content area, etc.

e) **Chart and Dashboard (every member):** Prepare three slides:

- Two slides with screenshots of charts and a summary of insights for each chart.
- One slide with a screenshot of the dashboard and a summary of insights from the dashboard.

f) **Recommendations (group):** Provide **not** more than 5 actionable recommendations. They must be based on the analysis and insights mentioned in the previous slides.

6. Presentation structure (25 minutes in total):

All members of a group must be present in the same timeslot. Where Group representative is stated, anyone in the group can represent.

Part	Time	Presenter	Tool(s)	What You Should Cover
1. Project Objective	1 minute	Group rep	PowerPoint	Briefly introduce your project objectives
2. Data Wrangling Walkthrough	3 minutes	Group rep	Python (code)	- Demonstrate how you cleaned or transformed the data using Python - Explain the main steps and why they were done
3. Dashboard & Storyboard Demonstration	5 minutes per member	Group members take turns	Python (code and demo)	- Demonstrate your dashboards and interactions - Walk through the charts and respective source code to explain how insights are derived - Highlight key visualisations and findings Note: Do not present the slides. Explain your charts and dashboard during the demonstration.
4. Recommendations & Wrap-Up	1 minute	Group rep	PowerPoint	Present clear recommendations based on insights
5. Q&A Session	5 minutes	Group members to answer	—	If the questions direct at individual work, the respective members will answer.

Section 3

Marking Scheme

Marks will be awarded to each student based on the following rubrics.

To score higher marks, you are encouraged to explore and experiment beyond the syllabus and demonstrate your independently-acquired skills via your deliverables / interview.

Component	Weightage
Group Component:	Total: 40 marks
1. Project Objective - State the objective(s) on analyzing the given datasets - Explain the purposes of the objectives	5
2. Dashboard Design Template Header Section: company logo and dashboard title	5

Component	Weightage
- Navigation Bar: Date Range, Categories, etc. - Main Content Area (3 charts) : Chart 1, Chart 2 and Chart 3	
3. Data Wrangling (Python) - Clean and preprocess the data - New fields are derived to facilitate chart creation	20
4. Storyboard (Plotly Dash) - All the dashboards are combined into one storyboard - Users can toggle between the dashboards	5
5. Data analysis and Presentaton (Powerpoint Slides) - Quality of Group slides - Coherence among all members' slides - Recommendations – based on all the charts and dashboards - Quality of presentation on group components	5
Individual Component:	Total: 60 marks
6. Charts and Dashboard (Python) a) Chart: Create three charts: one (1) Plotly Express chart and two (2) Plotly Graph Objects. All the charts must include interactive elements such as radio buttons, checkboxes, dropdowns, slides, etc. b) Dashboard: Create one interactive dashboard (Plotly Dash) that integrates interactive elements such as radio buttons and filters. c) Code quality & Documentation (e.g., modular code with comments) The following criteria applies to all charts and dashboard: - Technical complexity: demonstrate advanced use of libraries, data processing, and visualization techniques - User-friendliness: intuitive layout, easy navigation and clear labelling - Aesthetics & Creativity: Visually appealing design with thoughtful use of colors, layout, and chart types	a) 3 charts * 8 marks = 24 marks b) 1 dashboard: 10 marks c) Code Quality: 6 marks
7. Data analysis (Powerpoint Slides) - Useful insights: Well-explained, enlightening and able to highlight achievements or areas of concern clearly - Quality of Slides: Individual slides are clear, comprehensive, professional and consistent with the group slides	10
8. Presentation, Interview and Q & A session Effectiveness in presenting and answering questions	10

- End of assignment specifications -